Practice · printf

Each question lists its options; the right one is marked **Right answer**. *Feedback* is what students see after they answer. In a typed question the blank is in square brackets, with the accepted answers separated by `|`.

(mc) L1 · What %d is

In `printf("Age: %d\n", age);`, what is `%d`?

- Text that is printed as it is.
- **Right answer:** A hole that is replaced by the value of `age`.
- A comment.
- A mistake.

Feedback: `%d` is a hole. It is not printed. The value of the variable after the comma is put in its place, so the screen shows `Age: 20`.

(mc) L1 · The hole for a whole number

Which hole prints an `int`?

- **Right answer:** `%d`
- `%f`
- `%c`
- `%i` and `%d` are both wrong

Feedback: `%d` is for `int`. `%f` is for `double`, `%c` for `char`.

(mc) L1 · The hole for two decimals

Which hole prints a `double` with exactly two digits after the decimal point?

- `%f`
- `%2f`
- **Right answer:** `%.2f`
- `%d`

Feedback: `%.2f`: the dot and the 2 say "two digits after the point". Plain `%f` gives six. `%2f` has no dot and does not do what you want.

(mc) L2 · What plain %f shows

What does this print?

```
double p = 3.75;
printf("%f\n", p);
```

- `3.75`
- **Right answer:** `3.750000`
- `4`
- `3`

Feedback: Plain `%f` always shows six digits after the point. For `3.75` use `%.2f`.

(mc) L2 · Two holes

What does this print?

```
int a = 3;
int b = 4;
printf("%d plus %d\n", a, b);
```

- `a plus b`
- `%d plus %d`
- `4 plus 3`
- **Right answer:** `3 plus 4`

Feedback: The first variable fills the first hole, the second fills the second. So `3 plus 4`.

(mc) L2 · Wrong hole

What happens with this?

```
double p = 3.75;
printf("%d\n", p);
```

- It prints `3.75`.
- It prints `3`.
- **Right answer:** It prints a nonsense number, and with `-Wall` the compiler warns you.
- The program does not compile.

Feedback: `%d` is for `int`, but `p` is a `double`. The program compiles and prints rubbish. Compile with `-Wall` and you get a warning telling you the hole and the type do not match.

(fib) L3 · Print an int

Complete the line so it prints `Year: 2026` from the variable `year`: printf("Year: [%d]\n", year);

Feedback: `%d`, because `year` is an `int`.

(fib) L3 · Print with two decimals

Complete the line so it prints `Height: 1.80` from the `double` variable `h`: printf("Height: [%.2f]\n", h);

Feedback: `%.2f`: a `double`, shown with two digits after the point.

(fib) L3 · Print a character

Complete the line so it prints `Grade: A` from the `char` variable `g`: printf("Grade: [%c]\n", g);

Feedback: `%c` is the hole for one character.